MAT2040

Tutorial 12

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Question 1

Let $\mathbf{x}$ and $\mathbf{y}$ be vectors in an inner product space. Show that if $\mathbf{x} \perp \mathbf{y}$ then the distance between $\mathbf{x}$ and $\mathbf{y}$ is

$$\left(\|\mathbf{x}\|^2 + \|\mathbf{y}\|^2\right)^{1/2}$$

Solution 1

If $\langle \mathbf{x}, \mathbf{y} \rangle = 0$, then

$$\begin{aligned} \|\mathbf{x} - \mathbf{y}\|^2 &= \langle \mathbf{x} - \mathbf{y}, \mathbf{x} - \mathbf{y} \rangle \\ &= \langle \mathbf{x}, \mathbf{x} \rangle - 2\langle \mathbf{x}, \mathbf{y} \rangle + \langle \mathbf{y}, \mathbf{y} \rangle \\ &= \|\mathbf{x}\|^2 + \|\mathbf{y}\|^2 \end{aligned}$$

Therefore

$$\|\mathbf{x} - \mathbf{y}\| = \left(\|\mathbf{x}\|^2 + \|\mathbf{y}\|^2\right)^{1/2}$$

Question 2

Use equation $\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{i=1}^{n} x_i y_i w_i$ with weight vector $\mathbf{w} = \left(\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\right)^T$ to define an inner product for $\mathbb{R}^3$, and let $\mathbf{x} = (1, 1, 1)^T$ and $\mathbf{y} = (-5, 1, 3)^T$.

(a) Show that $\mathbf{x}$ and $\mathbf{y}$ are orthogonal with respect to this inner product.

(b) Compute the values of $\|\mathbf{x}\|$ and $\|\mathbf{y}\|$ with respect to this inner product.

Solution 2

(a) $\langle \mathbf{x}, \mathbf{y} \rangle = x_1 y_1 w_1 + x_2 y_2 w_2 + x_3 y_3 w_3 = 1 \cdot -5 \cdot \frac{1}{4} + 1 \cdot 1 \cdot \frac{1}{2} + 1 \cdot 3 \cdot \frac{1}{4} = 0$

Question 3

Let θ be a fixed real number and let

$$\mathbf{x}_1 = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix} \quad \text{and} \quad \mathbf{x}_2 = \begin{pmatrix} -\sin\theta \\ \cos\theta \end{pmatrix}$$

(a) Show that $\{\mathbf{x}_1, \mathbf{x}_2\}$ is an orthonormal basis for $\mathbb{R}^2$.
(b) Given a vector $\mathbf{y}$ in $\mathbb{R}^2$, write it as a linear combination $c_1\mathbf{x}_1 + c_2\mathbf{x}_2$.
(c) Verify that

$$c_1^2 + c_2^2 = \|\mathbf{y}\|^2 = y_1^2 + y_2^2$$

Solution 3

(a)

$$\mathbf{x}_1^T \mathbf{x}_1 = \cos^2\theta + \sin^2\theta = 1$$
$$\mathbf{x}_2^T \mathbf{x}_2 = (-\sin\theta)^2 + \cos^2\theta = 1$$
$$\mathbf{x}_1^T \mathbf{x}_2 = -\cos\theta\sin\theta + \sin\theta\cos\theta = 0$$

(c)

$$\begin{aligned} c_1^2 + c_2^2 =& (y_1\cos\theta + y_2\sin\theta)^2 + (-y_1\sin\theta + y_2\cos\theta)^2 \\ =& y_1^2\cos^2\theta + 2y_1y_2\sin\theta\cos\theta + y_2^2\sin^2\theta \\ & + y_1^2\sin^2\theta - 2y_1y_2\sin\theta\cos\theta + y_2^2\cos^2\theta \\ =& y_1^2 + y_2^2 \end{aligned}$$

Question 4

A. Let $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_k, \mathbf{x}_{k+1}, \ldots, \mathbf{x}_n\}$ be an orthonormal basis for an inner product space V. Let S_1 be the subspace of V spanned by $\mathbf{x}_1, \ldots, \mathbf{x}_k$, and let S_2 be the subspace spanned by $\mathbf{x}_{k+1}, \mathbf{x}_{k+2}, \ldots, \mathbf{x}_n$. Show that $S_1 \perp S_2$.

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B. Let $\mathbf{x}$ be an element of the inner product space V in Exercise 33, and let $\mathbf{p}_1$ and $\mathbf{p}_2$ be the projections of $\mathbf{x}$ onto S_1 and S_2, respectively. Show that
(a) $\mathbf{x} = \mathbf{p}_1 + \mathbf{p}_2$.
(b) if $\mathbf{x} \in S_1^{\perp}$, then $\mathbf{p}_1 = \mathbf{0}$ and hence $S_1^{\perp} = S_2$.

Solution 4

If $\mathbf{u} = c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + \cdots + c_k\mathbf{x}_k$ is an element of S_1 and $\mathbf{v} = c_{k+1}\mathbf{x}_{k+1} + c_{k+2}\mathbf{x}_{k+2} + \cdots + c_n\mathbf{x}_n$ is an element of S_2, then

$$\begin{aligned}\langle \mathbf{u}, \mathbf{v} \rangle &= \left\langle \sum_{i=1}^{k} c_i\mathbf{x}_i, \sum_{j=k+1}^{n} c_j\mathbf{x}_j \right\rangle \\ &= \sum_{k=1}^{k}\sum_{j=k+1}^{n} c_i c_j \langle \mathbf{x}_i, \mathbf{x}_j \rangle \\ &= 0\end{aligned}$$

Solution 4

(a) By Theorem 5.5.2,

$$\begin{aligned}\mathbf{x} &= \sum_{i=1}^{n} \langle \mathbf{x}, \mathbf{x}_i \rangle \, \mathbf{x}_i \\ &= \sum_{i=1}^{k} \langle \mathbf{x}, \mathbf{x}_i \rangle \, \mathbf{x}_i + \sum_{i=k+1}^{n} \langle \mathbf{x}, \mathbf{x}_i \rangle \, \mathbf{x}_i \\ &= \mathbf{p}_1 + \mathbf{p}_2\end{aligned}$$

(b) It follows from Exercise 31 that $S_2 \subset S_1^{\perp}$. On the other hand if $\mathbf{x} \in S_1^{\perp}$ then by part (a) $\mathbf{x} = \mathbf{p}_1 + \mathbf{p}_2$. Since $\mathbf{x} \in S_1^{\perp}, \langle \mathbf{x}, \mathbf{x}_i \rangle = 0$ for $i = 1, \ldots, k$. Thus $\mathbf{p}_1 = \mathbf{0}$ and $\mathbf{x} = \mathbf{p}_2 \in S_2$. Therefore $S_2 = S^{\perp}$.